

**EGR 3350-07**

**Technical Communications for Engineers and Computer Scientists**

**Ms. Alysoun Taylor-Hall**

## **Individual Portfolio**

Utsav Acharya

Computer Science

April 30, 2026

# **Table of Contents**

## **1. Mechanism Description**

- Revision Plan
- Revised Report
- Original Graded Assignment

## **2. Process Description**

- Reflection
- Original Graded Assignment

## **3. Recommendation Report**

- Reflection
- Original Graded Assignment

## **4. Group Project (Group 3)**

- Reflection
- Letter of Intent (LOI)
- Annotated Bibliography
- Proposal

# **1. Mechanism Description**

## **Revision Plan**

EGR 3350-07

Technical Communications for Engineers and Computer Scientists

Ms. Alysoun Taylor-Hall

Utsav Acharya | Computer Science | April 30, 2026

## **Revision Plan: Wireless Fast Phone Charger**

### **Overview of Original Feedback**

The Mechanism Description received a score of 6.5 out of 7.0 (A). The instructor identified several recurring issues: (1) overuse of contractions, which are inappropriate for technical writing; (2) imprecise pronoun use, particularly sentences beginning with 'it' or 'it's' where the subject should be explicitly named; (3) informal word choices and constructions that do not suit technical description; and (4) sentences flagged as 'unclear' or attempting to do 'too much at once.' The revision addresses each of these issues systematically across all sections.

### **Planned Changes by Section**

#### **Introduction**

The introduction will replace all contractions with full forms (e.g., 'It's' becoming 'It is') and refine the measurement of phrasing to read 'measuring approximately 10 cm in diameter and 0.8 cm in height. The pronoun 'its,' when used to open a clause about the charger's surface, will be clarified so the grammatical subject is the charger itself.

#### **Charging Surface**

This section received the most feedback. The instructor flagged the use of 'It's' (a contraction), the informal phrase 'so that the,' and noted that contractions are 'ok, but not for the subject of a sentence.' The revision rewrites every sentence using 'The charging surface is...' as the consistent grammatical subject, eliminates all contractions, and splits any compound sentence flagged as doing 'too much at once' into shorter, direct statements.

#### **Indicator Light Ring**

The phrase beginning with 'So it glows' was flagged as unclear and informal. The revision rewrites this to explicitly state that the indicator light ring illuminates in different colors



to communicate charging status, removing the informal connector and making the signaling mechanism precise.

### Protective Casing

The instructor struck through the informal description 'It's a tough plastic enclosure' and indicated cleaner phrasing. The revision adopts the instructor's suggestion: 'It is made of tough plastic and is approximately 9.5 cm wide and 0.8 cm high.' All remaining informal phrasing in this section is replaced.

### Power Input Port

The instructor recommended 'It is approximately' to replace the original phrasing, and flagged 'so that the user' as informal. The revision adopts the instructor's phrasing: 'The port is approximately 8 mm wide,' and replaces 'so that' constructions with precise technical alternatives such as 'allowing the user to...' or 'enabling repeated daily use without...'

### Conclusion

The instructor advised to 'use short, direct sentences' and avoid trying to do too much at once. The conclusion is rewritten as a sequence of clear, direct statements — each sentence naming one component and its function without embedding multiple ideas in a single clause.

## **General Style Changes Across All Sections**

The revision makes the following changes throughout: (1) eliminates all contractions, replacing them with full forms; (2) begins each component description with the full component name as the grammatical subject rather than a pronoun; (3) breaks any sentence flagged as 'unclear' or 'too much at once' into two shorter sentences; and (4) replaces informal connectors such as 'so' and 'so that' with precise technical language.

## **Revised Mechanism Description**

EGR 3350-07

Technical Communications for Engineers and Computer Scientists

Ms. Alysoun Taylor-Hall

## **Wireless Fast Phone Charger**

Utsav Acharya

Computer Science

April 30, 2026

## Introduction

A *wireless fast phone charger* is a power-transfer device that supplies electrical energy to a smartphone without a direct, wired connection. Energy transfers through electromagnetic induction when a compatible phone is placed on the charger's upper surface. The charger is circular, black, and constructed of rigid polymer plastic, measuring approximately 10 cm in diameter and 0.8 cm in height. The mechanism consists of four observable components: the charging surface, the indicator light ring, the protective casing, and the power input port.

## Discussion

Figure 1 shows the wireless fast phone charger and its primary components.

### Charging Surface

The *charging surface* is a circular support platform that stabilizes a smartphone during energy transfer. It is a flat, round platform approximately 10 cm across and 0.8 cm thick. The surface has a matte finish that provides friction, preventing the phone from sliding during charging. The flat shape positions the device consistently over the induction coil beneath, ensuring reliable energy transfer. Around the outer edge of the charging surface runs the indicator light ring.

### Indicator Light Ring

The *indicator light ring* is a visual signaling component that communicates charging status to the user. It has a diameter of approximately 0.2 cm and is constructed of translucent plastic. The indicator light ring illuminates in different colors depending on whether the device is actively charging. When a phone is charging, the ring emits a steady light; when no device is detected, the ring remains unlit. The indicator light ring encircles the charging surface and rests on top of the protective casing.

## **Protective Casing**

The *protective casing* is a structural enclosure that houses and supports the internal electrical components, including the induction coil and supporting electronics. It is made of tough plastic and measures approximately 9.5 cm wide and 0.8 cm high. The casing shields the internal components from physical impact and environmental exposure. Small rubber pads on the bottom surface of the casing prevent the charger from sliding on flat surfaces. The power input port is integrated into the side of the protective casing.

## **Power Input Port**

The *power input port* is a connector that enables the charger to receive power from an external source. The port is located on the side of the casing and is approximately 8 mm wide, accommodating a standard USB-C plug. The enclosure surrounding the port is constructed of durable material, allowing repeated daily connection and disconnection without degradation. Once connected, power flows directly into the induction system housed within the casing.

## **Conclusion**

A *wireless fast phone charger* is a power-transfer device that supplies electrical energy to a smartphone without a direct, wired connection. The charging surface holds the device in the correct position over the induction coil. The indicator light ring signals the current charging status to the user. The protective casing encloses and shields the internal electronics. The power input port connects the unit to an external power source. Together, these four components enable

fast, reliable wireless charging.

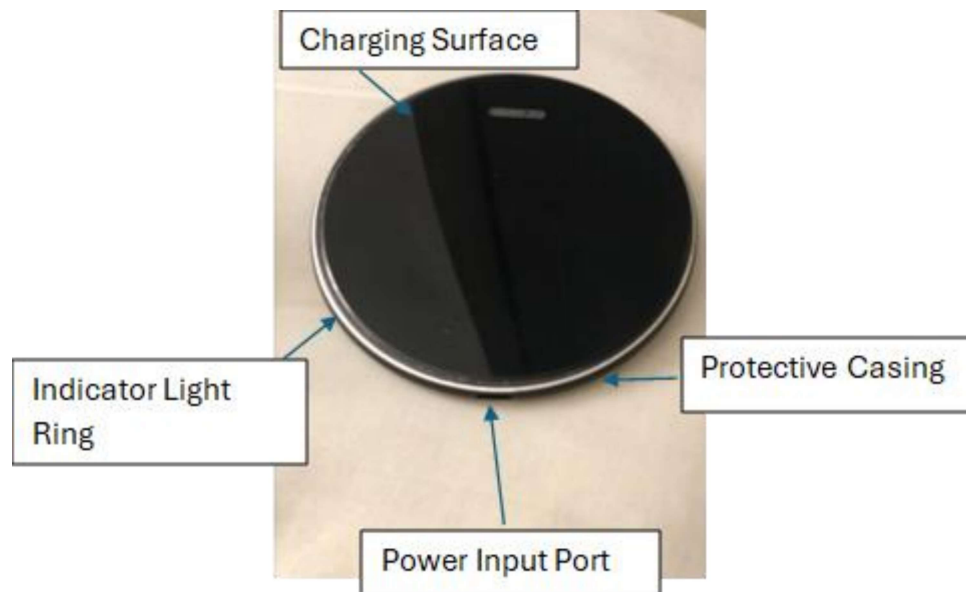


Figure 1. Wireless Fast Phone Charger, showing labeled components. Source: Author.

EGR 3350-07 ✓

Technical Communications for Engineers and Computer Scientists ✓

Ms. Alysoun Taylor-Hall ✓

Mechanism Description Final Draft ✓

6.5/7.0

A



Wireless Fast Phone Charger ✓

Utsav Acharya ✓

Computer Science ✓

February 10, 2026 ✓

## Introduction

*A wireless fast phone charger* is a power-transfer device that supplies electrical energy to a smartphone without a direct wired connection. Energy transfers through electromagnetic induction when a compatible phone is placed on its upper surface. The charger is circular, black, and constructed of rigid polymer plastic, <sup>the charger's</sup> measuring approximately 10 cm in diameter and 0.8 cm <sup>high</sup> in height. The mechanism consists of four observable components: the charging surface, the indicator light ring, the protective casing, and the power input port.

## Discussion

Figure 1 shows the wireless fast phone charger and its primary components.

### Charging Surface

*The charging surface* is a circular support platform that stabilizes a smartphone during energy transfer. <sup>It is</sup> ~~It is~~ a flat, round platform <sup>that is</sup> 10 cm across and 0.8 cm thick, ~~with~~ a matte finish that gives it a bit of grip <sup>so that the</sup> phone won't slide off. Its flat shape keeps the device in the right spot for consistent charging. Around this surface runs the indicator light ring.

### Indicator Light Ring

*The indicator light ring* is a visual signaling component that communicates charging status. It has a diameter of approximately 0.2 cm thick and <sup>is</sup> made of translucent plastic. <sup>is</sup> ~~It~~ glows in different colors depending on whether the phone's charging or not. Once the light turns on, <sup>unclear</sup> reflect right away if it's working. All of this sits on top of the protective casing.

### Protective Casing

*The protective casing* is a structural enclosure that houses and supports internal electrical components. ~~It's a tough plastic enclosure,~~ 9.5 cm wide and 0.8 cm high, <sup>This</sup> that protects the ~~It is made of tough plastic and is~~

induction coil and electronics. Small rubber pads on the bottom keep the charger from sliding around on the desk. The power input port is built into the side.

## Power Input Port

The power input port is a connector that enables the charger to get power from an outside source.

*It is approximately 8 mm wide.*  
The port <sup>is</sup> on the side of the casing, ready for a USB-C plug, <sup>so that the user</sup> about 8 mm wide. The enclosure is tough, <sup>the device is</sup> can plug and unplug it every day without wearing it out. Once connected, power flows directly into the induction system inside.

## Conclusion

A wireless fast phone charger is a power-transfer device that supplies electrical energy to a smartphone without a direct wired connection. The charging surface holds device in place, the indicator ring shows what's happening, the casing protects the inside, and the input port brings in power. Together, these pieces work seamlessly to help charge quickly and easily.

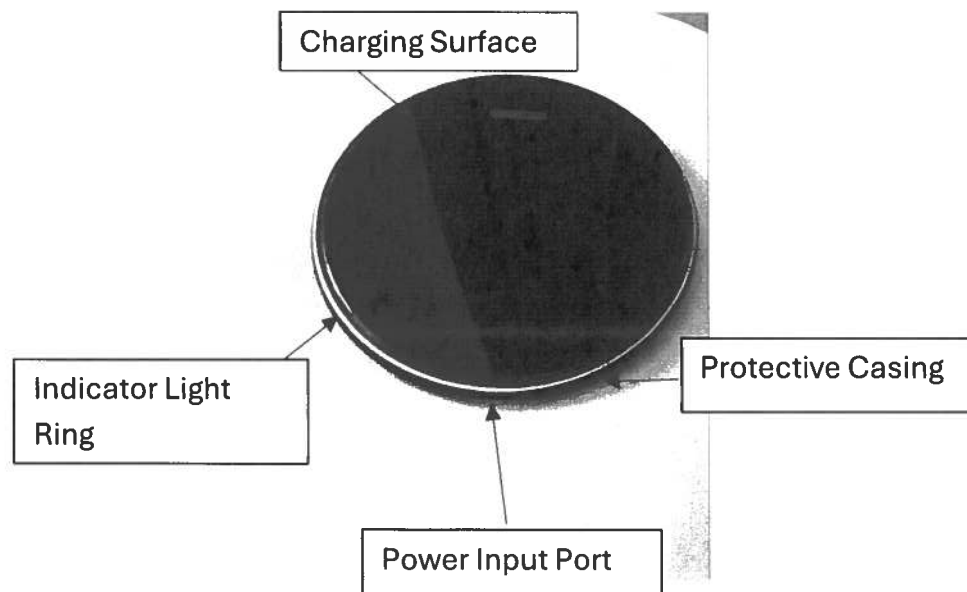


Figure 1. Wireless Fast Phone Charger, showing labeled components.

*Good work*  
*Be careful of leaving words out*  
*Use short, direct sentences — don't try to do too much at once*  
Source: Author.





## **2. Process Description**

### **Reflection**

EGR 3350-07

Technical Communications for Engineers and Computer Scientists

Ms. Alysoun Taylor-Hall

Utsav Acharya | Computer Science | April 30, 2026

## **Reflection: Two-Factor Authentication Verification Process**

Writing the process description for the two-factor authentication (2FA) verification process was a valuable exercise in structuring technical information for a general audience. Looking back on the assignment, I can identify both strengths in the paper and areas where my writing could have been more effective.

One aspect I am most satisfied with is the logical flow of the five-step sequence. By breaking the authentication process into clearly labeled stages: entry of primary credentials, system validation and code generation, transmission of the verification code, entry and verification of the secondary code, and access authorization, I provided readers with a clear roadmap before they encountered the details of each step. The instructor's comment that it was 'nicely done' confirmed that the overall structure was sound. The flowchart in Figure 1 also reinforced the written explanation effectively, giving readers a visual reference for the sequence.

However, reviewing the instructor's feedback reveals two areas for growth. First, the professor noted that the last two steps in the flowchart, 'User Enters Code' and 'Access Granted or Denied,' could be combined. This points to a broader issue in my writing: I was not always thinking critically about whether each step represented a meaningfully distinct stage in the process or whether I was artificially separating ideas that belong together. In technical writing, unnecessary divisions can create confusion rather than clarity.

Second, the feedback on the 'Entry and Verification of Secondary Code' section flagged imprecise word choice. The word 'either' was struck through and replaced with a more precise term. This reflects a tendency I noticed in my own writing: occasionally choosing words that feel

natural in everyday speech without verifying that they carry precise technical meaning. In a document describing a security process, precision is essential.

Overall, this assignment taught me that even a well-organized technical document benefits from a final critical review at both the structural and sentence level. I came away with a clearer understanding of how to evaluate whether divisions genuinely serve the reader and how to select language that is accurate and accessible. These lessons will shape how I approach all future technical writing.

6.7/7.0  
A

EGR 3350-07 ✓

Technical Communications for Engineers and Computer Scientists ✓

Ms. Alysoun Taylor-Hall ✓

Process Description ✓

Two-Factor Authentication Verification Process ✓

Utsav Acharya ✓

Computer Science ✓

February 24, 2026 ✓

# Introduction

*Two-factor authentication* is a security verification process that confirms a user's identity by requiring two separate forms of authentication credentials before granting access to a system or account. The purpose of this process is to enhance protection by decreasing the chances of unauthorized entry. The process occurs in five sequential steps: entry of primary credentials, system validation and code generation, transmission of the verification code, entry and verification of the secondary code, and access authorization.

## Discussion

Figure 1 illustrates the sequence of steps in the two-factor authentication process.

### Entry of Primary Credentials

*Entry of primary credentials* is an initiation step during which a user attempts to access a secure system by entering a username and password. These aspects are submitted through a login interface where the system compares the information with stored data in an authentication database. If the credentials are valid, the process advances to the second stage; if they are incorrect, access is denied. Once the login is validated, the process moves to system validation and code generation.

### System Validation and Code Generation

*System validation and code generation* are computational steps during which the system produces a temporary, randomly created verification code. The code remains active for a specific duration, typically between 30 and 60 seconds, to prevent the reuse of captured security codes. This serves as a secondary security measure in addition to password protection. After the code is successfully generated, it is ready for the transmission of the verification code.

### Transmission of Verification Code

*Transmission of verification code* is a delivery process during which the system sends the generated code to the user through an alternative authentication method. The system supports methods such as text messages, emails, or mobile authentication applications. The system validates the second authentication factor by sending it to a device that the authorized user currently possesses. The user must first receive the code before they can enter and verify their secondary code.

### **Entry and Verification of Secondary Code**

*Entry and verification of the secondary code* is a process of matching where the user enters the received code into the system for comparison. The authentication process continues when the submitted code matches the original generated value and the code remains valid. The system needs users to start a new login process when their information <sup>either</sup> becomes ~~either~~ invalid or expired<sup>s</sup>. The final step of the process requires users to receive access authorization after they have successfully completed the matching procedure.

### **Access Authorization**

*Access authorization* is a final granting step during which the system provides the user with full account access. This occurs only after both authentication factors, the password and the temporary code<sup>§</sup> have been successfully verified. This requirement of physical presence or possession of a secondary device ensures the security of the account.

### **Conclusion**

*Two-factor authentication* is a security verification process that confirms a user's identity by requiring two separate forms of authentication credentials before granting access to a system or account. The process starts with the entry of primary credentials and the system generation of a code; it involves the transmission of that code to the user, followed by the entry and verification

of the secondary code. Finally, the process concludes with access to authorization. By utilizing knowledge-based and possession-based methods, this process significantly decreases unauthorized access risk.

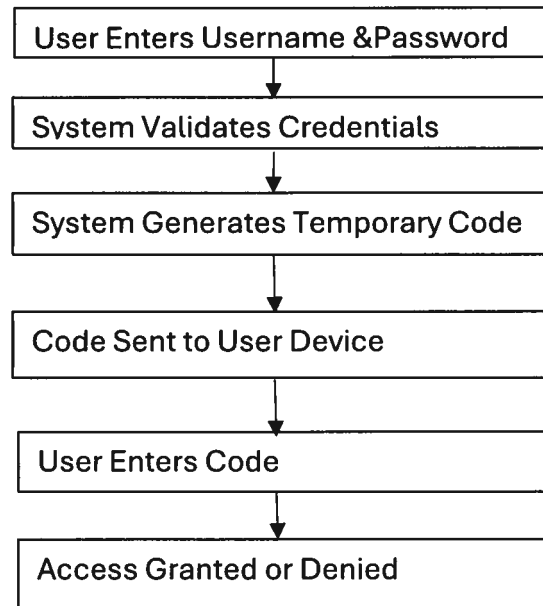


Figure 1. Two-Factor Authentication verification process. Source: Author.

Nicely done.  
The last 2 steps could be  
combined.



### **3. Recommendation Report**

#### **Reflection**

EGR 3350-07

Technical Communications for Engineers and Computer Scientists

Ms. Alysoun Taylor-Hall

Utsav Acharya | Computer Science | April 30, 2026

## **Reflection: Selecting the Best Password Manager for College Students**

The recommendation report comparing Bitwarden, 1Password, and LastPass for college student use was the most research-intensive writing assignment of the semester. Earning a score of 16 out of 16 was gratifying, and the instructor's written feedback was 'Very nicely done' and 'Well researched.' 'Good work' confirmed that the report was both technically sound and clearly presented. This reflection examines why the report succeeded and what habits I want to carry forward.

The report's greatest strength was its evidence-based structure. Each of the three evaluation criteria, security features, ease of use, and annual cost, followed an explanation data-interpretation pattern that guided the reader from context to evidence to conclusion before any recommendation was offered. This prevented the report from reading like an opinion piece and positioned the final recommendation as the logical outcome of the research. The three comparison tables reinforced this by giving readers a concise visual summary at each stage.

The selection of peer-reviewed and industry sources also contributed to the report's credibility. Drawing on IEEE conference papers about password manager usability and security, alongside official pricing pages from Bitwarden, 1Password, and LastPass, allowed me to ground every factual claim in a verifiable source. Providing six properly formatted IEEE-style references demonstrated that the research process was systematic rather than superficial.

If I were to revise this report, I would focus on one self-identified limitation: the scope was restricted to individual student plans, which meant I did not explore whether family or student-discount pricing might change the recommendation. A brief acknowledgment of this

boundary in the scope section would have made the report more transparent about the limits of its conclusions.

More broadly, this report reinforced that strong technical writing is as much about disciplined research as it is about clear prose. The process of locating credible sources, extracting relevant data, and presenting it in a form matched to the audience's needs is a skill I will continue to develop throughout my engineering career.

EGR 3350-07 ✓

Technical Communications for Engineers and Computer Scientists ✓

Ms. Alysoun Taylor-Hall ✓

**Recommendation Report** ✓

16/16  
A+

Selecting the Best Password Manager for College Students ✓

Utsav Acharya ✓

Computer Science ✓

March 17, 2026 ✓

# Introduction

## Purpose

The purpose of this report is to select and recommend the best password manager for college students.

## Problem

College students use multiple online services such as email, banking, learning platforms, and social media accounts. Many students reuse passwords or choose weak passwords due to difficulty remembering multiple credentials, creating significant cybersecurity risks, including unauthorized account access and identity theft. Research shows that password manager interaction problems with websites reduce adoption and lead users to rely on insecure workarounds [1]. A password manager improves security by storing and generating strong, unique passwords. Students require a solution that is affordable, secure, and easy to use across multiple devices.

Three password managers are being considered: Bitwarden, 1Password, and LastPass.

## Scope

This report compares Bitwarden, 1Password, and LastPass based on security features, ease of use, and annual cost. The report focuses on plans appropriate for individual student use.

## Discussion

### Security Features

### Explanation

Security evaluation refers to the ability of each password manager to protect stored credentials from unauthorized access. Key factors include encryption standards, zero-knowledge architecture, multi-factor authentication (MFA) support, and security audit history. Research comparing open-source password manager applications using ISO/IEC 25010 quality standards shows meaningful differences in how password managers protect user data [2]. The strength of a password manager depends on its cryptographic implementation, including AES-256 symmetric encryption, which is widely considered resistant to brute-force attacks [3]. ✓

## Data

Bitwarden uses AES-256 encryption, zero-knowledge architecture, and supports MFA including TOTP and YubiKey hardware keys [4]. Its open-source codebase is publicly available for independent security audits, and no major breaches have been reported. 1Password uses AES-256 encryption with an additional Secret Key for vault decryption and holds SOC 2 Type 2 certification [5]. LastPass uses AES-256 encryption and MFA but suffered a major data breach in 2022 in which encrypted vault data was exfiltrated by attackers [6]. Table 1 compares the security features of each password manager.

| Feature        | Bitwarden                | 1Password                       | LastPass           |
|----------------|--------------------------|---------------------------------|--------------------|
| Encryption     | AES-256 [4]              | AES-256 [5]                     | AES-256 [6]        |
| Architecture   | Zero-knowledge [4]       | Zero-knowledge + Secret Key [5] | Zero-knowledge [6] |
| MFA Support    | TOTP, email, YubiKey [4] | Yes [5]                         | Yes [6]            |
| Open Source    | Yes [4]                  | No [5]                          | No [6]             |
| Breach History | None reported [4]        | None reported [5]               | Yes - 2022 [6]     |
| Security Rank  | 2                        | 1                               | 3                  |

 ✓

Source: Author

## Interpretation

Bitwarden and 1Password both offer strong security architectures. Bitwarden's open-source model provides additional transparency and community oversight. 1Password's Secret Key provides a meaningful additional protection layer. LastPass ranks third due to the documented 2022 breach. Overall, 1Password ranks first for security, followed by Bitwarden, then LastPass.

## Ease of Use

Good

### Explanation

Ease of use evaluates how easily students can install, configure, and use the password manager in their daily workflow. Key factors include browser integration, mobile support, cross-platform compatibility, and user interface quality. Research demonstrates that interaction problems between password managers and websites significantly reduce adoption rates among users [1].

### Data

Bitwarden offers browser extensions for all major browsers, mobile apps for iOS and Android, and desktop clients for Windows, macOS, and Linux [4]. 1Password is widely recognized for its polished interface, browser extensions, mobile and desktop clients, and passkey storage support [5]. LastPass provides strong browser integration but has received mixed usability reviews following platform changes, including removal of free-tier multi-device support in 2021 [6].

### Interpretation

1Password provides the best overall user experience due to its polished interface. Bitwarden offers strong usability with full cross-platform compatibility. LastPass falls behind due to recent platform restrictions. 1Password ranks first, Bitwarden second, and LastPass third in usability.

## Annual Cost

Good

### Explanation

Cost evaluates the annual subscription price for individual students. For students with limited budgets, affordability is a significant factor in adoption. A solution that requires no payment for core functionality lowers the barrier to entry.

## Data

Bitwarden offers a free plan with unlimited password storage across unlimited devices and a premium plan costing approximately \$19.80 per year [4]. 1Password does not offer a free plan; its individual plan costs approximately \$35.88 per year [5]. LastPass offers a limited free plan restricted to one device type; its premium plan costs approximately \$36 per year [6]. Table 2 compares the annual cost of each password manager.

| Password Manager | Free Plan                      | Annual Premium Cost | Cost Rank |
|------------------|--------------------------------|---------------------|-----------|
| Bitwarden [4]    | Yes (unlimited devices)        | \$19.80/year        | 1         |
| 1Password [5]    | No                             | \$35.88/year        | 2         |
| LastPass [6]     | Yes (limited —one device type) | \$36/year           | 3         |

Source: Author

good

## Interpretation

Bitwarden provides the lowest cost, offering both a capable free tier and an affordable premium upgrade. 1Password and LastPass are significantly more expensive annually. For cost-conscious students, Bitwarden is the clear winner in this criterion.

## Conclusion

### Summary

This report evaluated Bitwarden, 1Password, and LastPass for college student use based on security features, ease of use, and annual cost. 1Password ranked highest in security and



usability, while Bitwarden offered the strongest overall value. Table 3 summarizes the overall comparison across all three criteria.

|           | Security | Ease of Use | Cost |
|-----------|----------|-------------|------|
| Bitwarden | 2        | 2           | 1    |
| 1Password | 1        | 1           | 2    |
| LastPass  | 3        | 3           | 3    |

*Source: Author; 1 = Best, 2 = Middle, 3 = Least*

## Conclusions

All three password managers provide AES-256 encryption and MFA support. They differ in security track record, usability quality, and cost. 1Password provides the strongest security and user experience but at the highest cost. Bitwarden delivers comparable security and reliable usability at a significantly lower cost. LastPass is the least recommended option due to its 2022 breach, declining usability reputation, and free-tier device restrictions.

## Recommendation

Bitwarden is recommended as the password management solution for college students. It provides strong AES-256 encryption, a zero-knowledge architecture, open-source transparency, and the lowest annual cost: free for basic use or approximately \$19.80 per year for premium access. For students who require the most polished interface and can accommodate a higher budget, 1Password is the preferred alternative.

## Contact

For more information, contact Utsav Acharya at [acharya.41@wright.edu](mailto:acharya.41@wright.edu)

*Very nicely done!*

## References

- [1] N. Huaman, S. Klivan, M. Oltrogge, Y. Acar and S. Fahl, “They Would Do Better if They Worked Together: The Case of Interaction Problems Between Password Managers and Websites,” 2021 IEEE Symposium on Security and Privacy (SP), San Francisco, CA, USA, 2021, pp. 1367–1381, doi: 10.1109/SP40001.2021.00094.
- [2] L. Nurdayenti and A. Amiruddin, “Comparative Analysis of Usability, Performance, and Security of Open-Source, Windows-Based Password Manager Applications Based on ISO/IEC 25010,” 2022 International Conference on Informatics, Multimedia, Cyber and Information System (ICIMCIS), Jakarta, Indonesia, 2022, pp. 178–183, doi: 10.1109/ICIMCIS56303.2022.10017543.
- [3] D. Ali, A. M. Tripathi and K. Saini, “A Study on Network Security and Cryptography,” 2024 1st International Conference on Advances in Computing, Communication and Networking (ICAC2N), Greater Noida, India, 2024, pp. 1–4, doi: 10.1109/ICAC2N63387.2024.10894984.
- [4] Bitwarden Inc., “Bitwarden Premium Pricing,” 2025. [Online]. Available: <https://bitwarden.com/pricing> [Accessed: Mar. 2026].
- [5] 1Password, “1Password Individual Plan Pricing,” AgileBits Inc., 2025. [Online]. Available: <https://1password.com/pricing> [Accessed: Mar. 2026].
- [6] LastPass, “LastPass Premium Pricing,” LastPass, 2025. [Online]. Available: <https://www.lastpass.com/pricing> [Accessed: Mar. 2026]

Well researched  
Good work



## **4. Group Project (Group 3)**

### **Reflection**

EGR 3350-07

Technical Communications for Engineers and Computer Scientists

Ms. Alysoun Taylor-Hall

Utsav Acharya | Computer Science | April 30, 2026

## **Reflection: Improving the Design of Floating Solar Farms**

Working on the Group 3 proposal to improve the efficiency and durability of floating solar farms in ocean environments was the most collaborative writing experience of the semester. The group consisted of five members: Carter Rockenbaugh, Daniel Evans, Erica Gilliam, Reese Harmeyer, and me, where Erica served as the team lead, helping coordinate the division of tasks and keeping the project on track.

From the beginning, our team divided the work according to each member's background and strengths. Carter focused on site selection and cost feasibility. Daniel handled the materials research, specifically the selection of high-density polyethylene (HDPE) and the anti-corrosive epoxy glass coating. Erica researched the mooring and anchoring system. My contribution centered on the software-based IoT monitoring and detection system, which aligned with my Computer Science background. I researched how sensors could continuously track panel performance and trigger the automated cleaning response when efficiency dropped below a set threshold. Reese works on cleaning robots.

The group worked well together overall. Meetings were productive, and we always left with clear next steps. The proposal came together better than I initially expected. The sections integrated smoothly because we agreed on a shared outline and consistent formatting before each person began writing independently. The LOI received a score of 7.7 out of 8.0 (A), and the instructor's feedback praised the work while asking us to clarify our exact scope further.

The most challenging part for me personally was bridging the technical depth of the monitoring system research with the need to keep the proposal accessible to a mixed engineering audience. I had to describe IoT sensor systems, voltage and current tracking, and automated alert

mechanisms in a way that Carter and Erica could follow without a software background. This required me to think carefully about audience adaptation, one of the core skills emphasized throughout the course.

Overall, this project demonstrated that group writing succeeds when members communicate consistently, respect each other's expertise, and agree on structure early. I came away with a stronger understanding of how collaborative technical documents are built and a greater appreciation for the role that clear division of labor plays in producing a cohesive final product.

7.7/8.0  
A

*Align*

To: Ms. Alysoun Taylor-Hall

From: Section 07, Group 3, Ustav Acharya, Daniel Evans, Erica Gilliam, Reese Harmeyer, Carter Rockenbaugh

Subject: Improving Floating Solar Farms in the Ocean

Date: April 4, 2026

## Purpose

We propose to improve the efficiency and durability of floating solar farms in the ocean to make them a more viable power solution. *Excellent*

## Summary

Solar power is a very well-known source of renewable energy. Unfortunately, solar panels often take up land that is needed for other things such as agriculture. But with the use of floating solar farms, that scarce land can be saved for more important things. Floating solar farms (FSFs) are emerging to be a viable option for large scale solar power production [1]. By putting solar panels into water, the solar panels efficiency is boosted due to the water keeping the panels cool.

Floating solar farms have only had major success on still water such as lakes or ponds and even fish farms. These farms have had trouble being placed successfully in the ocean. The idea becomes more complex due to the salt water and stronger waves causing faster wear and tear on the panels and floatation platforms that they sit on. The salt water also dirties the panels themselves and if the complete setup is not maintained properly, it will not be able to generate enough power [2].

*Avoid overpromising*

With \$250,000 our team plans to solve these issues to make floating farms more available for ocean use. Our team will do research on the best locations to install solar farms in the ocean. This will help with the placement of our anchoring and mooring system that will help to keep the floating panels in place and weaken the strength of incoming waves on the platforms. The team will also do research on a cleaning method to keep the panels clear of contaminants to ensure maximum power is being collected. This research will reduce long term cost of installing floating farms in the ocean as it will keep the farm in working order for longer.

## Qualifications

*Exact scope? Prototype? testing?*

Carter R. is a mechanical engineer who has experience in analyzing how systems perform technically and economically. Carter will be responsible for finding a suitable location and weighing the costs of making our solar farm better. Daniel E. is also a mechanical engineer with knowledge of mechanical properties of materials and structural designs. He will be responsible for selecting suitable material and floatation systems for the solar panels. Utsav Acharya is a Computer Science student with experience in software development using Java, Python, JavaScript, and PHP. He will be responsible for researching and developing the software-based dirty panel monitoring and detection system for the floating solar farm. Reese H. is a computer engineer with experience installing and working on electrical systems. Reese will be responsible for the development of the cleaning system that will work with a detections system for the

*good*

*Good work. Work on clarifying exact scope*

panels. Erica G. is an industrial engineer with experience in modeling and data analysis. Erica will be responsible for analyzing the materials to find the most suitable for this application.



## References

- [1] Sree, D. K. K., Law, A. W.-K., Pang, D. S. C., Tan, S. T., Wang, C. L., Kew, J. H., Seow, W. K., & Lim, V. H. (2022). Fluid-structural analysis of modular floating solar farms under wave motion. *Solar Energy*, 233, 161–181. <https://doi.org/10.1016/j.solener.2022.01.017>
- [2] Thangarasu, S. K., Gobinath, V. K., Shanmugam, A., Kumar, S. M., Shrinidhi, K. M., & Sanjay, S. (2024). Innovative Arduino-based solar panel cleaning system for enhanced energy efficiency. *2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT), Kamand, India*, 1–5. <https://doi.org/10.1109/ICCCNT61001.2024.10725904>



Font?

## Annotated Bibliography

Bru, J., Seland, T. S., Dai, J., & Jiang, Z. (2025). Life cycle cost analysis of an offshore floating

photovoltaic concept in the North Sea. *Renewable Energy*, 249, Article 122981.

<https://doi.org/10.1016/j.renene.2025.122981>

check for issue #

This journal article explains how much it costs to maintain offshore floating solar panels and how long they last before they need to be replaced. We will use this information to help design solar farms that are more cost efficient. This source was submitted by Carter Rockenbaugh. 1

Hegarty, A., Westbrook, G., Glynn, D., Murray, D., Omerdic, E., & Toal, D. (2019, April 15–18).

*A low-cost remote solar energy monitoring system for a buoyed IoT ocean observation platform* [Conference session]. 2019 IEEE World Forum on Internet of

Things (WF-IoT), Limerick, Ireland. <https://doi.org/10.1109/WF-IoT.2019.8767311>

Conference  
This paper discusses a solar monitoring system designed for ocean-based platforms. It shows how solar systems can be monitored remotely in harsh marine environments. <sup>we</sup> will use this source to directly connect monitoring systems to floating solar farms, showing how the system works in ocean conditions. This source was submitted by Utsav Acharya. 2

Huang, L., Elzaabalawy, H., Sarhaan, M., Sherif, A., Ding, H., Ou, B., Yang, D., & Cerik, B. C.

(2025). Developing reliable floating solar systems on seas: A review. *Ocean*

*Engineering*, 322, Article 120525. <https://doi.org/10.1016/j.oceaneng.2025.120525>

This journal article explains why installing solar panels in the ocean are more difficult to install. The article explains various designs and materials that help

issue #?

improve the stability and efficiency of floating solar farms. We plan to use this article when deciding which design is best for floating solar farms. This source was submitted by Daniel Evans. 1

Jayaraman, T., Vijayakumar, M., Jayakumar, T., Murali, A., Karthikeyan, B., & Suriya, T.

(2024). A conceptual design of remote controlled solar panel cleaning robot. *2024 5th International Conference on Smart Electronics and Communication (ICOSEC)*, Trichy, India, 387–391. <https://doi.org/10.1109/ICOSEC61587.2024.10722191>

This conference paper describes the possible design for a solar panel cleaning robot. We plan to use this article to help design the cleaning system for the floating solar farms. This source was submitted by Reese Harmeyer. 1

Kekre, A., & Gawre, S. K. (2017). Solar photovoltaic remote monitoring system using IoT.

*International Conference on Recent Innovations in Signal Processing and Embedded Systems (RISE)*, Bhopal, India, 508–512.

<https://doi.org/10.1109/RISE.2017.8378227>

This conference paper explains how IoT-based systems monitor solar panel performance using sensors and microcontrollers. It collects real-time data such as voltage, current, and temperature and sends it to a remote system for analysis and fault detection. <sup>we</sup> will use this source to explain how monitoring systems detect performance drops and identify when solar panels are not working efficiently. This source was submitted by Utsav Acharya. 2

Nair, A., Huang, L., & Verdin, P. G. (2025). Computational fluid dynamics and potential flow

Xiong, Y., Liu, H., Xu, J., & Chen, H. (2025). Mechanical properties of offshore floating photovoltaic structural coating/carbon steel system in a marine environment.

*Ocean Engineering*, 323, Article 120585.

check for issue #

<https://doi.org/10.1016/j.oceaneng.2025.120585>

This journal article looks at the mechanical properties of coated carbon steel in offshore solar farms. It explains why anti-corrosive coatings are important to maintain structural integrity. We plan to use this source when looking at materials.

This source was submitted by Daniel Evans. 2

Zeng, F., Bi, C., Sree, D., Huang, G., Zhang, N., & Law, A. W.-K. (2023). An adaptive barrier mooring system for coastal floating solar farms. *Applied Energy*, 348, Article 121618.

<https://doi.org/10.1016/j.apenergy.2023.121618>

Issue #?

Describe the contents

We plan to use this journal article to give us more information about building barriers beneath the water. This helps to keep the waves from knocking on the floating platforms and will extend the life of the solar panels. This source was submitted by Erica Gilliam. 2

Good mix of sources  
most properly identified  
A couple lack descriptions

modelling techniques for floating photovoltaic systems: A systematic review.

*Symmetry*, 17(9), 1508. <https://doi.org/10.3390/sym17091508>

*Describe the contents*

We can use this journal article to tell us more about the research that has already been done on the effects of waves on solar panels. This research will help us get a better understanding of what we need to improve on what has already been done.

This source was submitted by Erica Gilliam. *1*

Thangarasu, S. K., Gobinath, V. K., Shanmugam, A., Kumar, S. M., Shrinidhi, K. M., & Sanjay,

S. (2024). Innovative Arduino-based solar panel cleaning system for enhanced energy efficiency. *2024 15th International Conference on Computing*

*Communication and Networking Technologies (ICCCNT)*, Kamand, India, 1–5.

<https://doi.org/10.1109/ICCCNT61001.2024.10725904>

This conference paper is about a cleaning system for solar panels that implements sprinkler systems controlled by an Arduino. We will use this to further understand how to create our own solar panel cleaning system. This source was submitted by Reese Harmeyer. *2*

Wen, Y., et al. (2025). The role of offshore wind and solar PV resources in global low-carbon

transition. *Science Advances*, 11(11), Article eadx5580.

<https://doi.org/10.1126/sciadv.adx5580>

This journal article is about the future of offshore clean energy as the world is transitioning to using green energy sources. We will use this information help make offshore solar panels more feasible for the future. This source was submitted by Carter Rockenbaugh. *2*

Course: EGR 3350, Section 07

Technical Communications for Engineers and Computer Scientists

Instructor: Ms. Alysoun Taylor-Hall

Assignment: RFP

Title: Floating Solar Farms in the Ocean

Names: Reese Harmeyer, Carter Rockenbaugh, Daniel Evans, Utsav

Acharya, Erica Gilliam

Date: 4/23/2026

# Abstract

This proposal will present a new design to improve the efficiency and lifespan of floating solar farms in ocean environments. Floating solar farms offer many advantages to land based solar farms which include: reducing land usage, lower installation costs, and increased energy efficiency. However, there are challenges in the ocean environment which cause problems as to how well offshore farms work. The biggest challenges are corrosion, dirty panel surfaces, and the harsh conditions of the ocean. All these challenges significantly reduce the system's performance and durability.

To combat these challenges, this project proposes a design prototype that integrates three main components: specialized materials, robust mooring system, and an automated monitoring and cleaning system. The design includes high-density polyethylene (HDPE) with an anti-corrosive epoxy glass coating to protect the panels from the saltwater. A flexible mooring system with anchors, rubber chains and wave barriers will keep the solar panel platform stable while simultaneously reducing structural damage from the motion of the water. The biggest innovation for this proposal is the IoT-based monitoring and cleaning system. Sensors will constantly track the solar panels' performance to find any reduced efficiency due to dirt, salt, algae and any other debris. When a panel's performance drops below a given threshold, the detection system will send the panel cleaning robot to the affected area. This system reduces the need for manual maintenance, optimizes the solar energy collected, and extends the lifespan of the panels.

Key Terms: Floating solar farm, Solar panel efficiency, Automated cleaning system, Energy optimization, Offshore solar power, Maintenance reduction

# Introduction



## Purpose

The purpose of this proposal is to improve the efficiency of floating solar farms by designing a sustainable prototype.

## Background

There has been a strong interest in developing and using cleaner energy sources. One common method of creating cleaner energy is by converting sunlight into electrical energy using solar panels. Solar panels are typically installed on land. This, however, takes up available land space. This is why floating solar farms are being developed by installing solar panels on bodies of water. According to Nedelcu et al. (2026), solar panels on water ultimately save land for things such as agriculture, housing, and infrastructure. Installing solar panels on water also reduces cost due to the installation being cheaper than purchasing land and installing solar panels (Nedelcu et al., 2026).

Floating solar farms also offer a better performance compared to land-based systems. According to Huang et al. (2025), floating solar panels are approximately 5% more efficient compared to land-based systems due to the surrounding water cooling the solar panels down. Floating solar farms are also able to absorb more sunlight than land-based systems due to the sunlight reflecting off of the surrounding water (Huang et al., 2025).

The challenge of designing floating solar panels is that they must be able to withstand environmental conditions, especially ocean conditions. While floating solar farms already have success in calm waters, the installation and maintenance become challenging when placing them in ocean environments. Factors such as saltwater causing corrosion, debris accumulation, strong winds, and strong waves can damage the system and reduce the system's efficiency (Huang et al., 2025; Thangarasu et al., 2024).

## Scope

This proposal addresses the design of floating solar panels and how materials, the mooring system, and a cleaning system which will improve the system's overall efficiency and viability in the ocean. A prototype cleaning robot will be created.

## Discussion

### Approach

The cost of offshore solar farms is influenced by several key factors, including the need to construct durable floating platforms, select corrosion-resistant materials, and implement reliable anchoring systems capable of withstanding ocean conditions. In addition, power transmission infrastructure must be developed to efficiently deliver energy back to shore. Within this project, the most significant cost will come from the development of the automated solar panel cleaning system. Because this component is new and requires extensive research, testing, and refinement, it represents a major portion of the initial investment. However, this investment is essential, as maintaining clean panels is critical for maximizing efficiency and ensuring consistent energy output over time.

Just as significant to costs is site selection. The strategic selection of a location is vital to ensuring optimal performance and feasibility. The most attractive sites for offshore solar power generation include locations where there are high levels of solar radiation, relatively calm bodies of water, and proximity to energy infrastructure facilities. Nearshore and coastal locations provide the perfect balance by cutting down on transmission expenses yet still ensuring consistent operation. Furthermore, this type of site selection also avoids potential problems

associated with land use issues, which makes offshore solar power generation highly beneficial in areas that are highly urbanized or densely populated.

The monitoring and cleaning system will improve floating solar farm efficiency by detecting when panels become dirty and triggering a cleaning response. Solar panels in ocean environments are exposed to salt water, algae, and debris, which accumulate on panel surfaces and reduce energy output by 10-25% (Mani & Pillai, 2010). To address this, an IoT-based software monitoring system will be installed on the floating platform to continuously track energy output from each panel. When the system detects a significant drop in efficiency, it will send an alert to operators indicating that cleaning is required.

The monitoring system will use sensors to collect real-time data including voltage, current, and power output from the solar panels (Kekre & Gawre, 2017). This data will be transmitted remotely, allowing operators to monitor farm performance without physical access to the offshore platform. When efficiency drops below a set threshold, the system will trigger an automated cleaning mechanism or notify maintenance teams. This approach reduces manual labor, prevents long-term efficiency loss, and extends the operational lifespan of the floating solar farm.

There needs to be a cleaning system for when the panels get dirty as this can significantly decrease their performance. The cleaning needs to be gentle to preserve the life span of the panels as aggressive or excessive cleaning could cause damage to the panels.

We propose to build a prototype robot with the following features: water jets, spinning brushes, wheels or tracks, and a control interface. The water jets will be fed by a pump using fresh water and will spray the dirty panels that the detection system determined. The robot will then use spinning circular brushes to gently clear the debris from the effected panels. We will do

the necessary research to determine the best way for the robot to move around using either wheels or tracks. A motor will be necessary to drive either system.

The materials used to design the floating solar farm are important to increase the system's overall viability over time. According to Xiong et al. (2025), the selected materials need to withstand the harsh ocean conditions such as saltwater exposure, UV rays, strong winds, and strong waves. These conditions are important to consider because they reduce the system's lifespan and durability, while increasing the risk of corrosion (Xiong et al., 2025). To address these conditions, the proposed design will consist of high-density polyethylene (HDPE) and anti-corrosive modified epoxy glass coating.

According to Nedelcu et al. (2026), HDPE is the most common material used for inshore floating solar farms. HDPE is a durable and buoyant material that can withstand UV rays and variable temperatures. However, this material should be combined with an anti-corrosive coating when designing an offshore system (Nedelcu et al. 2026). The anti-corrosive coating that will be used in this design is a modified epoxy glass coating. According to Xiong et al. (2025), the modified epoxy glass coating improves the lifespan to 18 months, nearly double of most anti-corrosive coatings. It also has an impedance modulus of  $1e6 \Omega\text{cm}^2$ , which indicates strong resistance to corrosion (Xiong et al., 2025). This coating can be reapplied every 18 months to continually protect the farm.

Floating solar farms need to be kept in place and above water after they are set up. It is important that the panels cannot drift very far from where they are supposed to be and don't get covered by water if the water level rises. If the floating panels have more freedom to move this can lead to micro-cracks in the modules which causes loss in durability, less buoyancy, and

endangers the safety of the solar system (Zeng et al., 2023). To avoid these problems a well-developed mooring system is needed to prevent any unnecessary damage to the panels.

We propose a mooring system that includes an anchor, rubber chains, and wave barriers. The anchor will be placed on the sea floor and allow for the chains to be attached to something. Having the mooring chains be made of rubber allows for better tension and are not as taught as steel. These chains need to have some freedom to allow for small movements in the platforms the panels are sitting on. A wave barrier would be placed just below the platforms. The barriers would dilute the power of incoming waves. This protects the platforms and will help to extend their lives.

## Result

At the end of the contract term, this project will deliver an improved design for floating solar farms. This design will include corrosion-resistant materials, a mooring system capable of withstanding harsh ocean conditions, a detection system that determines when the solar panels need to be cleaned, and a cleaning system that automatically cleans the solar panels. The final outcome of this project will produce more sustainable energy by increasing the overall efficiency, durability, and lifespan of floating solar farms. These improvements benefit the Road Runner Research Foundation by reducing maintenance costs, increasing system durability, and supporting the public by producing cleaner energy.

## Statement of Work

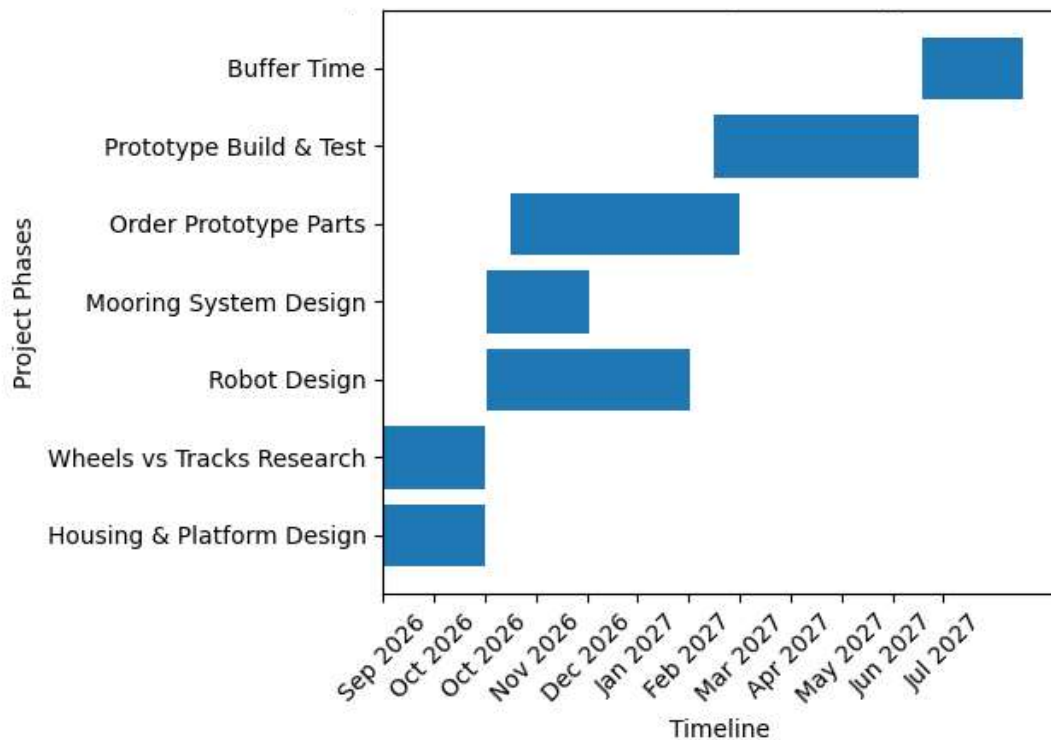
To achieve the goals of this proposal, the following tasks will be accomplished:

- **Task 1:** Design floating platform for the solar panels, material selection and structural layout for offshore location. (*2 months*)

- **Task 2:** Conduct research to determine the movement system of the cleaning robot, choosing either wheels or track systems. *(2 months, concurrent with Task 1)*
- **Task 3:** Design the automated solar panel cleaning robot. *(4 months)*
- **Task 4:** Design the mooring system, including anchors, rubber lines, and wave barrier. *(2 months, concurrent with Task 3)*
- **Task 5:** gather materials and components required for the robot prototype. *(concurrent with Task 3)*
- **Task 6:** Build and test the prototype, including performance evaluation. *(4 months)*
- **Task 7:** Perform final evaluation and adjust system to ensure reliability and completion of the project. *(2 months)*

## Timeline

First, we will start designing the housing and platform that will contain the solar panels in September of 2026. This will take a maximum of 2 months to complete. During this same time research will be done on whether the cleaning robot should use wheels or tracks. Next, we will start designing the cleaning robot based on whether wheels or a track system was chosen. This process will take a maximum of 4 months. At the same time we will start designing the mooring system for the solar farm. This process will take a maximum of 2 months to complete. While the cleaning robot is being designed, we will start to order parts for the prototype so that we can start working on it when the design is finished. When the design is finished for the cleaning robot we will start building and testing the prototype. This will take a maximum of 4 months to complete. We will have an excess of two months just in case the timeline doesn't go as predicted.



(OpenAI, 2026)

## Conclusion

### Summary

The proposed designs will make solar farms more viable and cost effective for ocean conditions. Our research will deliver: A design for solar farm housing using ocean resistant materials; A design for a coastal based mooring system that will keep the panels secure; A design for a detection system that will sense which panels need to be cleaned; A prototype cleaning robot. As land space becomes less available, these design innovations will help solar panels in the ocean output enough power to become a viable option for renewable energy.

# Utsav Acharya

Fairborn, OH | (937) -313-2397 | [acharya.41@wright.edu](mailto:acharya.41@wright.edu) | <https://utsavacharya.com.np/>

I am a Computer Science student with a strong academic record and hands-on project experience. I am a fast learner who enjoys and thrives in fast-paced environments. I'm dedicated to continued improvement, personally and professionally.

## Education

**Wright State University, Dayton, OH**      **May 2024-present**

*Bachelor of Science in Computer Science*

Coursework for this degree, which is focused on software development, systems, and mathematics which includes Computer Organization (Assembly), Discrete Math, Calculus I & II, Operating System, Computer Science I & II (Java), Statistics for Computer Scientists, Elementary Linear Algebra, Logic for Comp Scientist, Intro to Cyber Security, Data Struc & Algorithms, Tech Comm for EGR & CS.

## Experience

**Wright State University, Math Learning Center**

The Math Learning Center is part of Wright State University's Student Success Center, offering free peer tutoring services to support students across a range of mathematics courses. In my current role, I provide one-on-one and group tutoring to help students succeed in their coursework, apply CRLA Level I certified instructional techniques to improve learning comprehension, and adapt my tutoring style to diverse student learning needs while maintaining an encouraging environment.

*Math Tutor*      *January 2025-present*

## Projects

**Hospital Management System**

**2022 Oxford Secondary School**

**Hackathon**

As team lead, I developed a centralized platform to manage hospital workflows, patient records, and administrative scheduling. The system was designed to streamline medical operations, with proposed future enhancements including e-prescriptions and automated doctor-matching for faster patient care. The project earned fourth place out of 19 competing teams.

**Dual Authentication System**  
**2025**

**Wright Hackathon**

As part of a 24-hour hackathon challenge, I collaborated with a team to build a secure dual authentication system integrating a mobile app and web interface to protect user data. The system implemented access-control features ensuring identity verification before accessing critical data, demonstrating rapid prototyping, innovative problem-solving, and strong teamwork. The project was awarded Best Collaboration Team at the Make It Wright Hackathon 2025.



## Technical Skills

Languages: Java, Python, C, SQL, JavaScript, PHP, Shell, Assembly, QBASIC

Web Development: HTML, CSS, React, Tailwind CSS

Tools: Git, GitHub, VS Code, Linux, MS Office, IntelliJ, Replit, DevC++, CLion

Soft Skills: Tutoring, Collaboration, Communication, Problem Solving, Leadership

## Awards and Recognition

CRLA Level I Certified Tutor, Wright State University MLC — 2025

Technology First 2025 Scholar

Recognized for Creativity & Collaboration — Make It Wright Hackathon 2025

Fourth Place Hackathon Ranking (out of 19 teams) 2022 Oxford Secondary School Hackathon

## Leadership & Community Engagement

Volunteer - Blood Donation Programs (Red Cross & Lions Club); led blood donation camp organization during high school.

School Quiz Team Representative, Grades 7-10 - competed in regional inter-school quiz competitions.

Science Exhibition Participant - created a prototype device to convert air pollution into usable acidic compounds for cosmetic and industrial uses.

# Daniel Evans

Dayton, OH | (937) 896-2276 | evansd161@gmail.com | *U.S. Citizen*

Current Mechanical Engineering student at Wright State University ready to apply problem-solving skills and technical skills. Looking to contribute to engineering projects while gaining valuable experience in the field.

## Education

### **Wright State University, Dayton, OH     Aug. 2024–Present**

#### *Bachelor of Science in Mechanical Engineering*

Mechanical engineering coursework with a focus on high-level math, physics, and engineering analysis. Relevant coursework includes SolidWorks, Matlab, Statics, and Properties of Materials.  
GPA: 3.934

### **Edison State Community College, Piqua, OH     Aug. 2022–May 2024**

#### *STEM Coursework*

Completed science and engineering coursework that built on my experience at Graham High School taking Honors and CCP courses.  
GPA: 3.90

## Projects

### **Matlab Application Design     May 2025**

Designed and developed four interactive applications using Matlab App Designer. Collaborated with a partner to design, test, and troubleshoot code, ensuring functionality across all applications.

### **Cardboard Bridge Design     Dec. 2024**

Designed, built, and tested a cardboard bridge with a partner, using SolidWorks for simulation and evaluating its strength and load capacity.

### **3D-Printing     July 2024**

Took part in a 3D printing project, gaining an understanding of the design process. Designed and created 3D-printed models using SolidWorks while learning design optimization.

## Technical Skills

SolidWorks (Certified SolidWorks Associate – Dec. 2024)

LTspice, Matlab, R

## Awards and Recognition

Recipient of Mercy Health's Give for Good Scholarship

Recipient of the 4-H Family and Consumer Sciences Scholarship

Wright State University Dean's List (3 semesters)

Edison State Community College Dean's List (1 semester)

# Erica Gilliam

(937) 679-1091 | charerica43@gmail.com | Dayton, OH | U.S. Citizen

I am a systems-focused professional with a work history that illustrates my commitment to learning and growth. I am a fast learner who enjoys and thrives in fast-paced environments. I'm dedicated to continued improvement, personally and professionally.

## Education

### Wright State University, Dayton, OH

Aug. 2025–present

#### *Bachelor of Science in Industrial and Systems Engineering*

Coursework for this degree, which is focused on improving the efficiency and quality of systems, is centered on high-level math, physics, and statistics. Classes taken to date include optimization, human fundamentals, advanced statistics, Six Sigma, Solid Works, Python, Matlab, and R.

GPA 3.7

### Sinclair College, Dayton, OH

Aug. 2022–Aug. 2025

#### *STEM Coursework*

Classes taken in science and engineering built on my experience at Kettering High School taking Honors and AP courses in their engineering program.

## Experience

### The TSP Solution, LLC, Dayton, OH

TSP Solution is a contracted delivery service provider (DSP) exclusive to Amazon at their Kettering, Ohio warehouse. In my current role, I manage a fleet of vans to support assigned routes, which vary daily and widely (from 35–75); monitor and adjust routes and/or drivers throughout the day to maintain Amazon's targeted performance; track vehicle maintenance and file all affiliated reporting; and investigate damage and other incident reports and work with the reporting parties toward a positive outcome.

#### *Fleet Manager*

April 2025–present

#### *Team Lead*

Oct. 2024–April 2025

#### *Delivery Driver*

April 2021–Oct. 2024

### The Kroger Co., Dayton OH

This Cincinnati-based grocery retailer operates more than 2,700 stores through the U.S., many of which include pharmacies and additional retail. One of the largest retailers in the country, they are a leader in developing best practices in inventory and in-store technologies.

#### *Assistant Drug/General Merchandise Manager*

May 2020–April 2021

#### *Grocery Clerk*

March 2019–May 2020

#### *Courtesy Clerk*

June 2017–March 2019

## Technical Skills

Six Sigma Green Belt, Solid Work, Python, Matlab, R

## Awards and Recognition

Recipient of the Wright State University Rike Associate Degree Transfer Scholarship, which recognizes academic performance

Sinclair College Dean's List for 6 of 8 quarters

# Reese Harmeyer

Miamisburg, OH (937) 581-5516 | rharmeyer2004@gmail.com | U.S. Citizen

I am a computer engineering student that is currently studying at Wright State. I am very good at adapting to different environments and can also learn complex processes very quickly. I have good work ethic, and I do things to the best of my ability.

## Education

### **Wright State University, Dayton, OH**

**Aug. 2025–present**

#### *Bachelor of Science Computer Engineering*

Coursework for this degree, which is focused on understanding digital and electrical systems as well as digital logic. It is centered on high-level math, computer science, and statistics. Classes taken to date include devices and circuits, digital circuit design, computer organization, and technical communications. I currently have a 4.0 GPA.

### **Sinclair College, Dayton, OH**

**Aug. 2022–May. 2024**

#### *Engineering Coursework*

I took gen ed courses like psychology and anthropology as well as the basic math and science courses that engineers need a framework to build from. I took AP Calculus at Carlisle High School which allowed me to take some extra courses.

## Experience

### **Camp Chautauqua Inc., Miamisburg, OH**

I currently work as a maintenance assistant. My main responsibility is to handle any sort of electrical project that the camp is doing whether we are doing new construction, putting up new lights, ballast bypassing, or repairing a new issue. When there isn't electric work to be done, I do various other tasks such as: maintaining equipment, framing, and plumbing.

#### *Maintenance Assistant*

*August 2024–present*

#### *Intern*

*May 2022–June 2022*

#### *Grill Manager*

*May 2021–August 2021*

### **Kings Electric., Lebanon OH**

I worked at Kings Electric for about 4 months as a sort of summer internship. I learned electrical code, how to bend conduit, and how to use a hammer drill to work with concrete.

#### *Apprentice Electrician*

*May 2023–August 2023*

## Technical Skills

C++, Java, Matlab, VHDL, Rockwell Software

## Awards and Recognition

Associates Transfer Scholarship

# Carter Rockenbaugh

| Dayton OH | (937) 578-3383 | carterrockenbaugh@gmail.com | U.S. Citizen |

I am a systems-focused professional with a work history that illustrates my commitment to learning and growth. I am a fast learner who enjoys and thrives in fast-paced environments. I'm dedicated to continued improvement, personally and professionally.

## Education

**Wright State University, Dayton, OH** Aug. 2025–present

*Bachelor of Science Mechanical Engineering*

Coursework for this degree, which is focused on the design, analysis, and improvement of mechanical systems, is centered on high-level math, physics, and engineering principles. Classes taken to date include statics, dynamics, thermodynamics, fluid mechanics, materials science, SolidWorks, Python, and MATLAB.

## Experience

**Walmart Supercenter, Urbana OH**

Walmart is a multinational retail corporation offering grocery and general merchandise services, including Online Grocery Pickup (OGP). In this role, I supported daily OGP operations by picking and staging customer orders, ensuring accuracy and quality; assisted with dispensing orders to customers; and collaborated with team members to maintain efficient workflow during high-volume periods.

*Associate*

*November 2022–May 2024*

**Hilton Garden Inn, Beavercreek OH**

Hilton Garden Inn is a full-service hotel brand focused on guest comfort and reliability. In this role, I performed routine maintenance and repairs across hotel facilities, troubleshooted mechanical and electrical issues, and responded to guest service requests to ensure a safe, functional, and comfortable environment.

*Maintenance Technician* January 2025 – July 2025

**Hotel Ardent by Hilton**

Hotel Ardent is a modern hospitality brand focused on delivering high-quality guest experience. In this role, I performed routine maintenance and repairs, troubleshooted mechanical and electrical issues, and responded to service requests to ensure a safe, functional, and well-maintained environment.

*Engineer* August 2025 – Present

## Technical Skills

Solid Work, MATLAB

## Awards and Recognition

AFROTC Scholarship Recipient; Recognizes a mix of academic ability, leadership potential, and overall character



## Facilities and Equipment

We will be partnering with Wright State University. They will provide an adequate space for our research and development of a prototype cleaning robot.

We will need the following resources:

- 2 SolidWorks Licenses
- 5 Desktop computers capable of running SolidWorks and other intense programs
- 1 Arduino
- 1 Electric motor
- 2 Radial brushes
- 1 Water pump
- 2 Water jets
- 1 Battery
- Miscellaneous supplies for the cleaning robot prototype

## Budget and Justification

Our proposed budget is Two hundred fifty thousand dollars; this would be for research of the floating solar farms and the prototype cleaning robot. A large portion of this budget would be allocated to the development of the cleaning robot, this would include research, design in 3d modeling, and prototype fabrication of the robot. These tests would ensure the system can reliably operate in harsh weather and marine conditions, the success of this robot in the testing and development phase are paramount. Specifically, the cleaning robot will need the following: An Arduino, an electric motor, radial brushes, a water pump, water jets, and a battery. The Arduino will cost \$22.80 (Audino, n.d.)

Additionally, we would need to use funds to pay the wages of hired individuals. The projected positions needed would be: 1 computer scientist, 1 computer engineer, 1 industrial engineer, and 2 mechanical engineers, who would be responsible for the design, construction and testing of the equipment and systems. They will each be paid \$20 per hour. A smaller portion of the fund would be used to acquire the unknown materials that will be needed for the cleaning robot. We will also have to pay for the design tools we use; such as SolidWorks which costs \$2320 per license (Dassault Systèmes, 2025.) Computers that can run intense software will also be required and will cost \$2999 per computer (HP Development Company, 2026.)

The remainder of our budget will be an emergency fund in case of design modifications or unforeseen occurrences that would hinder our progress. Overall, this budget would allow ample funds to ensure the success of our project, from initial to final testing, and support flexibility for unforeseen challenges.

| Category  | Item                                             | Number     | Subtotal | Total       |
|-----------|--------------------------------------------------|------------|----------|-------------|
| Personnel | Computer Scientist Salary, \$20/hour             | 2080 hours | \$41600  | \$41600.00  |
| Personnel | Computer Engineer Salary, \$20/hour              | 2080 hours | \$41600  | \$83200.00  |
| Personnel | Industrial Engineer Salary, \$20/hour            | 2080 hours | \$41600  | \$124800.00 |
| Personnel | 2 Mechanical Engineers Salary, \$20/hour         | 4160 hours | \$83200  | \$20800.00  |
| Equipment | SolidWorks (Dassault Systèmes, 2025)             | 2          | \$5640   | \$213640.00 |
| Equipment | Desktop Computers (HP Development Company, 2026) | 5          | \$14995  | \$228635.00 |
| Equipment | Arduino (Arduino, n.d.)                          | 1          | \$22.80  | \$228657.80 |
| Supplies  | Electric Motor                                   | 1          | \$500    | \$229157.80 |
| Supplies  | Radial brushes                                   | 2          | \$500    | \$229657.80 |

|                 |                                        |     |         |             |
|-----------------|----------------------------------------|-----|---------|-------------|
| <b>Supplies</b> | Water Pump                             | 1   | \$100   | \$229757.80 |
| <b>Supplies</b> | Water Jets                             | 2   | \$75    | \$229832.80 |
| <b>Supplies</b> | Battery                                | 1   | \$300   | \$230132.80 |
| <b>Supplies</b> | Miscellaneous Cleaning Robot Materials | N/a | \$15000 | \$245132.80 |

## References

- Arduino. (n.d.). *Arduino Micro*. Retrieved April 23, 2026, from <https://store-usa.arduino.cc/products/arduino-micro>
- Dassault Systèmes. (2025, May). *SOLIDWORKS design plans & pricing*. <https://www.solidworks.com/how-to-buy/solidworks-design-plans-pricing>
- HP Development Company. (2026). *HP Z2 Tower G1i desktop workstation with 3 yr warranty & Wolf Pro Security*. <https://www.hp.com/us-en/shop/pdp/hp-z2-tower-g1i-workstation-desktop-pc-wolf-pro-security-edition-p-bp8m3ua-aba-1>
- Huang, L., Elzaabalawy, H., Sarhaan, M., Sherif, A., Ding, H., Ou, B., Yang, D., & Cerik, B. C. (2025). Developing reliable floating solar systems on seas: A review. *Ocean Engineering*, 322, Article 120525. <https://doi.org/10.1016/j.oceaneng.2025.120525>
- Kekre, A., & Gawre, S. K. (2017). Solar photovoltaic remote monitoring system using IoT. *International Conference on Recent Innovations in Signal Processing and Embedded Systems (RISE)*, Bhopal, India, 508–512. <https://doi.org/10.1109/RISE.2017.8378227>
- Mani, M., & Pillai, R. (2010). Impact of dust on solar photovoltaic (PV) performance: Research status, challenges and recommendations. *Renewable and Sustainable Energy Reviews*, 14(9), 3124–3131. <https://doi.org/10.1016/j.rser.2010.07.065>
- Nedelcu, A.-T., Faităr, C., Voicu, I., & Panaitescu, M. (2026). Floating photovoltaic systems—Energy performance and environmental challenges in sustainable development. *Sustainability*, 18(5), Article 2588. <https://doi.org/10.3390/su18052588>
- OpenAI. (2026). *Gantt chart of solar panel cleaning robot project timeline* [AI-generated image]. ChatGPT. <https://chatgpt.com/>
- Thangarasu, S. K., Gobinath, V. K., Shanmugam, A., Kumar, S. M., Shrinidhi, K. M., & Sanjay,

S. (2024). Innovative Arduino-based solar panel cleaning system for enhanced energy efficiency. *2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT)*, Kamand, India, 1–5.

<https://doi.org/10.1109/ICCCNT61001.2024.10725904>

Xiong, Y., Liu, H., Xu, J., & Chen, H. (2025). Mechanical properties of offshore floating photovoltaic structural coating/carbon steel system in a marine environment. *Ocean Engineering*, 323, Article 120585. <https://doi.org/10.1016/j.oceaneng.2025.120585>

Zeng, F., Bi, C., Sree, D., Huang, G., Zhang, N., & Law, A. W.-K. (2023). An adaptive barrier mooring system for coastal floating solar farms. *Applied Energy*, 348, Article 121618. <https://doi.org/10.1016/j.apenergy.2023.121618>